

# Scoliosis



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# Session Objectives

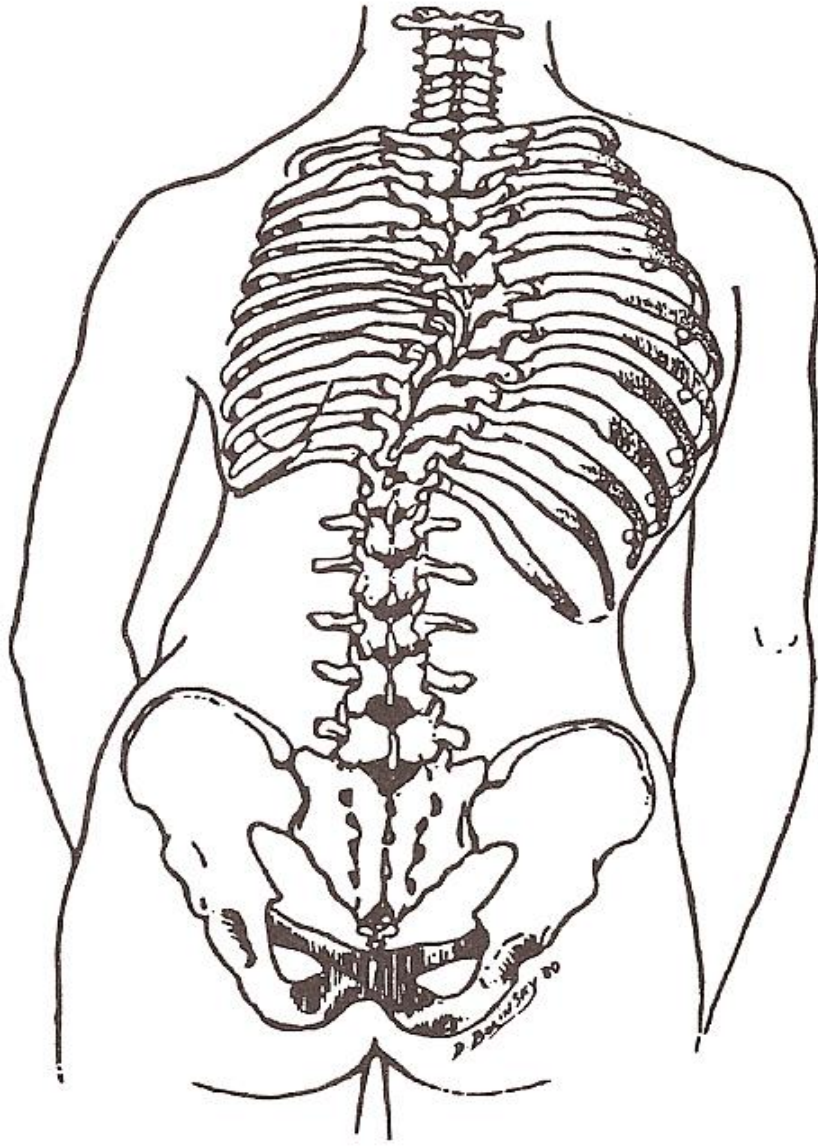
1. the students will be able to learn the various types and classifications of scoliosis.
2. the students will be able to learn the standard radiological diagnostic method for assessing severity of scoliosis.
3. the students will be able to learn the treatment options for scoliosis.
4. the students will be able to learn the treatment outcomes for scoliosis.

# Scoliosis

- Scoliosis – is a **description** of a structural alteration, or alterations, that can occur with several conditions, or can occur “idiopathically.”

# Scoliosis

- Scoliosis – defined as a lateral curvature of the spine in the coronal plane, usually accompanied by rotation of variable amounts.
- Therefore a.k.a. rotoscoliosis, since both the coronal and transverse planes are often involved.



### Truncal asymmetry in scoliosis

Scoliosis may be indicated by differences in the height of the shoulders or scapulae and asymmetry of the waistline, all of which are depicted in the above photograph.

*Courtesy of Susan A Scherl, MD.*

Scherl, Susan, MD, et. al. Adolescent idiopathic scoliosis: Clinical features, evaluation, and diagnosis, UpToDate: literature review current through June 2026, topic last updated June 18, 2026.

LINK: [https://login.nyit.idm.oclc.org/login?url=https://www.uptodate.com/nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search\\_result&selectedTitle=2%7E11&usage\\_type=default&display\\_rank=2](https://login.nyit.idm.oclc.org/login?url=https://www.uptodate.com/nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search_result&selectedTitle=2%7E11&usage_type=default&display_rank=2)

# Scoliosis

- Lateral group curves equal to or greater than 10 degrees (when the Cobb Method of measurement is used) are called **scoliosis**.
- Spinal curves less than 10 degrees are termed **spinal asymmetry**. These curves have no long-term clinical significance.
  - E.g. Type I group curves

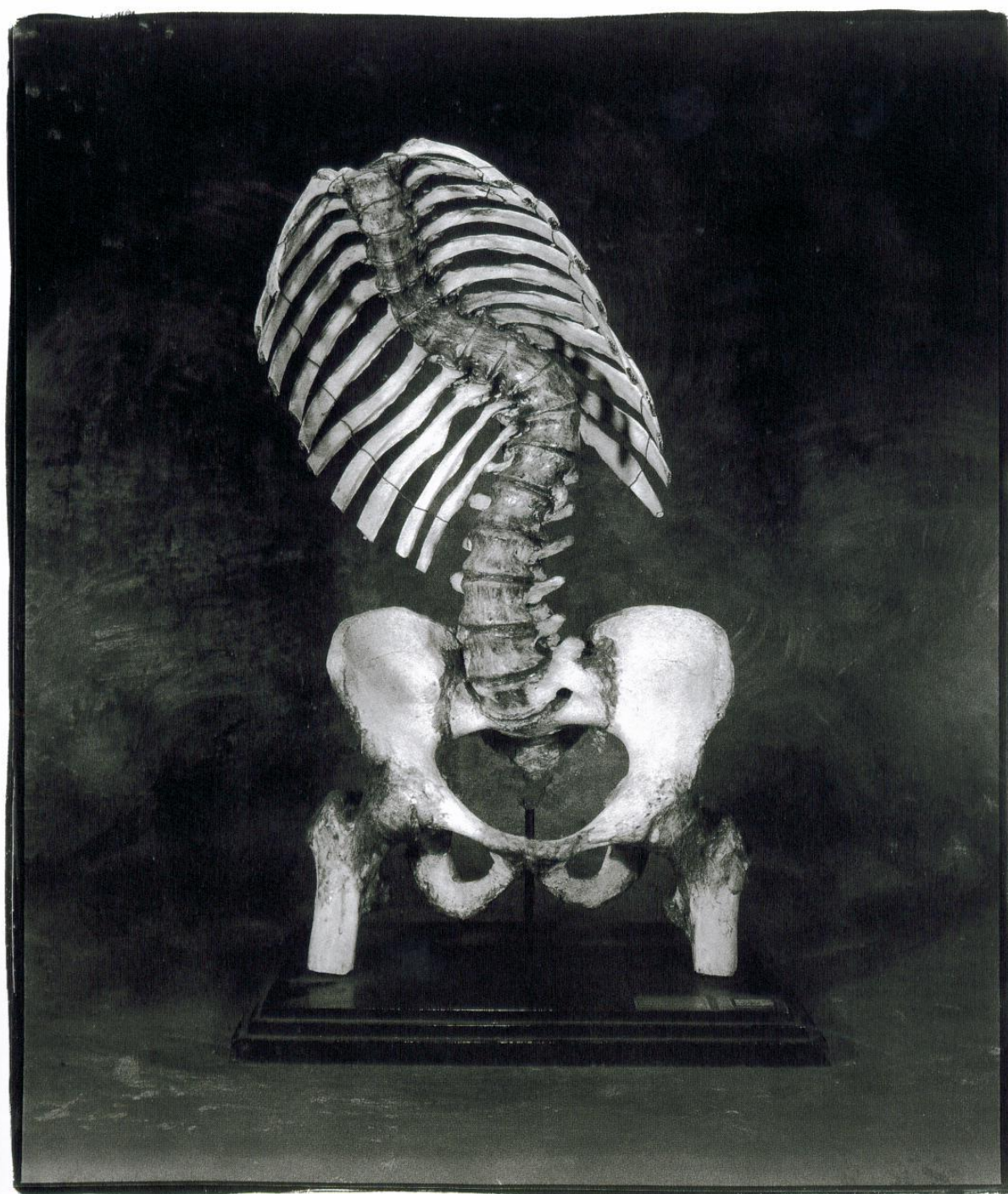
# Scoliosis

- Scoliosis is named for the side of the **convexity** of the lateral/coronal plane curve.
- The **apical vertebra** is the one that is the most deviated and rotated away from the midline.

# Scoliosis

- Sagittal plane curves are kyphotic or lordotic curves.
- If three planes are involved, there could be a kyphoscoliotic curve.
- Significant deformity can occur during times of rapid growth, when the curve will most likely progress and can cause cardiopulmonary compromise.





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# Scoliosis - Classifications

- Several ways to classify scoliosis including:
  - Reversibility
  - Severity
  - Cause
  - Location

# Scoliosis - Classifications

- **Reversibility**
  - **Functional or structural – Structural** curves do not go away with rotational, side-bending, or forward bending motions. **Functional** curves, however, do go away with these motions.
  - Can have patient bend forward until maximal rib hump is visible, then ask patient to swing upper body right then left. Look for reduction of the rib hump with this action.

# Change from functional to structural

- Progressive soft tissue shortening
- Body changes
- Disc degeneration
- Fibrosis of ligaments & fascia
- Decreased active & passive ROM

# Scoliosis - Classifications

- **Severity**
  - **No Scoliosis**
  - **Mild Scoliosis** – 10 to 15 degrees
  - **Moderate Scoliosis** – 20 to 45 degrees
  - **Severe Scoliosis** – more than 50 degrees, and will affect systemic function as well.
    - Thoracic curves **>50 degrees** affect **respiratory** function.
    - Thoracic curves **>75 degrees** affect **cardiovascular** function

# Scoliosis - Classifications

- Cause
  - Idiopathic – 70% – 90% of all scoliotic curves (most common). As more is learned, some in this category may be reclassified as coming from *sacral base or cranial base unleveling*, or from *sagittal plane biomechanics*.
    - Infantile
    - Juvenile
    - Adolescent
  - Congenital
  - Acquired
  - Neuromuscular

# Scoliosis - Classifications

- **Infantile Idiopathic Scoliosis** –
  - Possibly due to supine and slight side-lying positioning, allowing gravity to cause plastic deformation of the immature thorax. Ideal treatment is prevention, by lying the kids prone (opposite of back to sleep recommendations)
  - Treatment with well fitting orthoses/ casting esp. in early period of skeletal growth.



# Scoliosis - Classifications

- Important difference between **early** and **late onset** scoliosis:
  - Any child with a significant scoliosis **before age 5** can develop **cardiorespiratory failure or disabling dyspnea**.
  - **Reason:** the scoliosis decreases the available space for growth of the lungs.
  - **Normally,** there is a tenfold increase of the alveoli from infancy to 4 years of age, continuing up to 8 years of age.



# Scoliosis - Classifications

- **Juvenile Idiopathic Scoliosis**
  - Between 4 and 10 years of age.
  - More progressive than adolescent idiopathic scoliosis
  - <20 degrees – observation with examination and PA x-rays every 4 to 6 months.
  - With evidence of progression (5-7 degrees), bracing is indicated.
  - Variable success of nonoperative treatment.
  - Crankshaft phenomenon – use anterior and posterior fusion to avoid this.
  - Subcutaneous Harrington Instrumentation to decrease amount of shortening (avg. = 4.5 cm if fusion had been done)

# Scoliosis - Classifications

- **Adolescent Idiopathic Scoliosis (AIS)**
  - Hereditary predisposition with multifactorial causes. Association with hypokyphosis, possible CNS defect, increased calmodulin, etc.
  - + genetic contribution supported by high concordance of the incidence and severity of scoliosis seen in monozygotic twins, as compared with dizygotic twins.
  - Patients with AIS also often have a sibling or parent with scoliosis.
  - Genetic loci for AIS mapped to chromosomes 17 and 19, with an unclear inheritance pattern.

# Scoliosis - Classifications

- Adolescent Idiopathic Scoliosis, cont.
  - Progression associated with:
    - Girls > boys;
    - premenarchal;
    - Risser sign of 0;
    - Double > single curves;
    - Thoracic > lumbar curves;
    - severe curves.
  - Measure height of patient standing and sitting

# Scoliosis - Classifications

- **Congenital** - second most common after idiopathic. 75% are progressive. May be associated with other congenital defects (cardiac, genitourinary)
  - Congenital vertebral defect(s):
    - Hemivertebrae
    - Wedge vertebrae
    - Failure of segmentation
      - Unilateral (bar)
      - Bilateral (fusion)
      - Fused ribs

# Scoliosis - Classifications

- **Acquired** –
  - for example: from osteomalacia, sciatic irritability, psoas syndrome, healed leg fracture, from a hip prosthesis, due to irradiation/inflammation, etc.

# Scoliosis - Classifications

## **Neuromuscular:**

**most common causes**

1. Leg length discrepancy
2. Neurofibromatosis

**Neuromuscular: curve disappears with forward flexion or lying down:**

- Leg length discrepancy
- Primary postural scoliosis

# Scoliosis - Classifications

## **(FYI) Neuromuscular:**

### **Neuropathy –**

- Cerebral palsy
- Diastematomyelia
- Friedrich ataxia
- Myelomeningocele
- Spinal cord injury
- Syringomyelia
- Tethered spinal cord

## **(FYI) Neuromuscular:**

### **Mesenchymal origin –**

- Congenital laxity of the joints
- Ehlers-Danlos syndrome
- Homocystinuria
- Marfan syndrome

# Scoliosis - Classifications

## **(FYI)** Neuromuscular: Trauma

- Direct vertebral trauma
- Extravertebral trauma
  - Severe trunk burns
  - Thoracic surgery
- Irradiation

## **(FYI)** Neuromuscular: Tumors

- Intraspinal
  - Extramedullary (neurofibroma)
  - Intramedullary (astrocytoma)
- Vertebral
  - Osteoid osteoma
  - Osteoblastoma



# Scoliosis - Classifications

## **(FYI) Neuromuscular: Myopathy**

- Arthrogryposis
- Cleidocranial dysostosis
- Duchenne muscular dystrophy
- Dysautonomia
- Facioscapulohumeral dystrophy
- Hyperphosphatasia
- Hypervitaminosis A
- Hypothyroidism
- Juvenile rheumatoid arthritis
- Limb-girdle muscular dystrophy
- Mucopolysaccharidoses
- Nemaline myopathy

# Scoliosis - Classifications

## **(FYI)** Neuromuscular: Miscellaneous

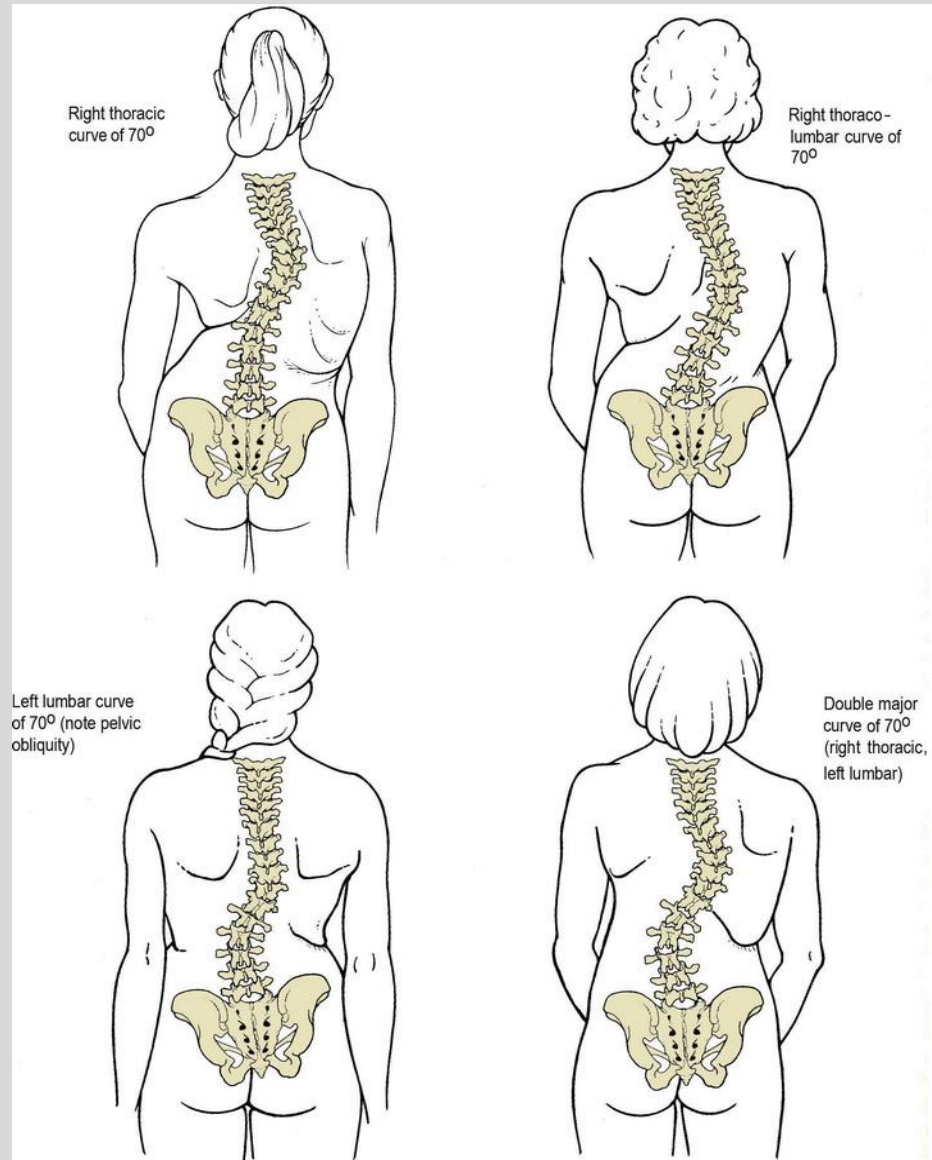
- Achondroplasia
- Charcot-Marie Tooth disease
- Juvenile spinal muscle atrophy
- Klippel-Feil syndrome
- Osteogenesis imperfecta
- Poliomyelitis
- Rickets
- Schueurmann disease
- Sprengel deformity
- Vertebral body infection

# Scoliosis - Classifications

- **Location**

- **Double Major Scoliosis** – most frequent is thoracic and lumbar; balanced curves; crossover regions subject to degeneration.
- **Single Thoracic Scoliosis** – second most common; if progressive, can affect heart and lungs; quite noticeable.
- **Single Lumbar Scoliosis** – third most common; associated with arthritic change.
- **Junctional Thoracolumbar Scoliosis** – not a common scoliosis; a longer curve; overstresses the spine; arthritic changes.

# Scoliosis - Classifications



# Scoliosis - Presentation

- **Children** – often asymptomatic; check between ages 10 to 15 when rapid bone growth is occurring, asymmetry may be noted.
- **Adolescents** – often asymptomatic; clothing doesn't fit well, asymmetry may be noted.
- **Adults** – symptoms of arthritis; backache; neck ache; headache; chest pain; organ dysfunction.

# Scoliosis - Presentation

- Patient may present to pediatrician, because the school nurse noted an abnormality during a scoliosis school screening.
  - Pediatrician should be ready to evaluate the patient for scoliosis, whether noted incidentally, discovered through a school screening program, or when the patient or parents express concern about scoliosis.
  - A spinal examination is important especially around age 10, just before the growth spurt of puberty.

# Scoliosis - Evaluation

- Exclude any underlying etiology
- Assess the magnitude of the curve
- Assess the risk for progression
- History:
  - Age of onset, rate of progression, any features of pain, neuromuscular complaints, shortness of breath, family history of scoliosis, current pubertal stage, onset of menarche (girls), any limb trauma history, infection, arthritis.

# Scoliosis - Evaluation

- Do a careful physical examination with spinal examination
- Inspection:
  - Patient's height – plot on gender and age specific graph
  - Stage of puberty (Tanner stage)
  - Neurologic examination
    - Normal walking, toe and heel walking, deep squatting, hop on each leg
    - Include abdominal reflex



# Scoliosis - Evaluation

- Do a careful physical examination with spinal examination
- Inspection:
  - Examination of the skin
    - Any café-au-lait spots and axillary freckling?
    - Any vascular, hypopigmented lesions or a patch or hair overlying the spine?
  - Any skin or joint laxity? Any high arches (pes cavus) or hammer/claw toes?

# Scoliosis - Evaluation

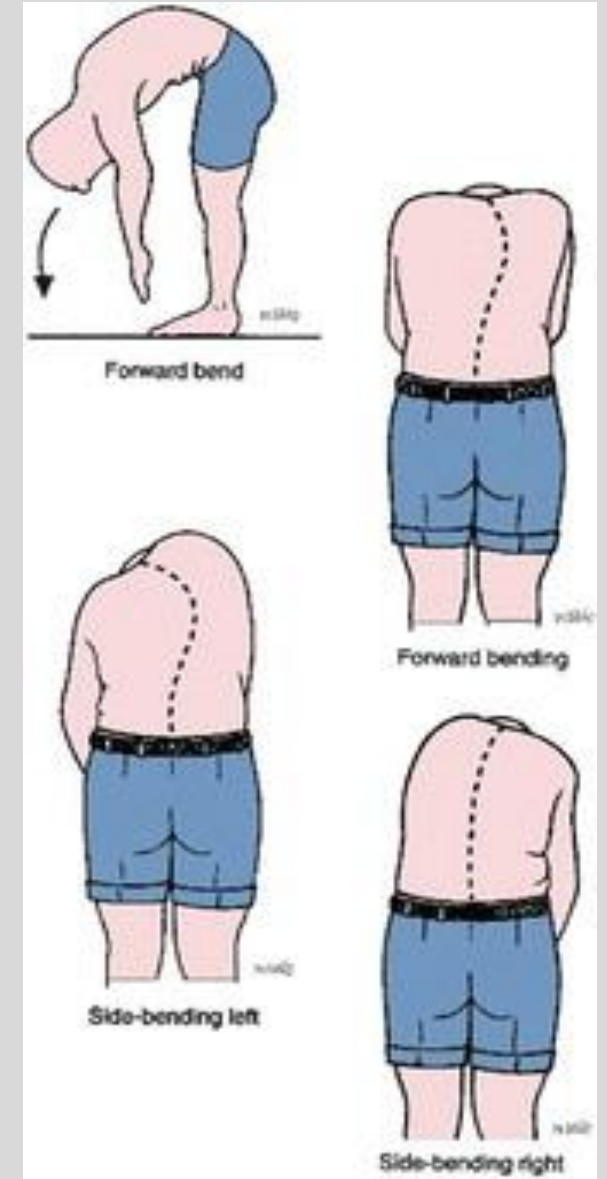
- Do a careful physical examination with spinal examination
  - Observe how arms are hanging by the body
  - Check levelness of occiput, shoulders, iliac crests, PSIS, PIIS, and greater trochanters.
  - Measurement of leg length
  - **Patient forward bending** (a.k.a. **the Adams forward bend test**) to observe for **Rib Hump**. Does it go away with sidebending to the side of the rib hump?
  - Run fingers along spinous processes.
  - Any lower extremity dysfunctions?

# Scoliosis - Evaluation

## Forward bending test for scoliosis.

Patient bends forward with knees extended and then side bends left and then right while physician observes posteriorly for accentuation or improvement of the spinal curve.

Functional curves improve or resolve *with side bending, whereas* structural curves do not.



(Illustration courtesy of William A. Kuchera, DO, FAAO, with permission).

From: Chapter 22 Osteopathic History and Physical Examination, *Figure 22.11*, *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research*, 4e, 2018, LINK: <https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-1wwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209544145&bookid=2582#209544208>

# Scoliosis - Evaluation



Figure 26.9, *Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research*, 4e, 2018, LINK:  
<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209545277&bookid=2582#209545302>

**A. Posterior view, checking for curvature in the spine.**

**B. Posterior view, patient standing and bent forward at the cervical spine level with chin to chest to assess for scoliosis in the upper thoracic and cervical regions.**

**C. Partial forward bent position while standing to assess for scoliotic curves in the thoracic and lumbar spinal regions.**

**D. Full forward bent position while standing to assess for scoliosis in the thoracic and lumbar spinal regions.**

*(Photos courtesy of Western University of Health Sciences, Pomona, CA, with permission.)*

# Scoliosis - Evaluation

- Do a careful physical examination with spinal examination
  - Use of a **scoliometer**:
    - Essentially is a carpenter's level
    - **Measures the angle of trunk rotation**
    - How to use:
      - The patient is forward bent, for the Adams forward bend test. Then, run the scoliometer along the patient's spine from cephalad to caudad. If a rib hump is present, the scoliometer ball will **deviate** away from the midline of the apparatus.
  - **Scoliometer degrees are NOT equivalent to the Cobb angle degrees.**
    - **General guideline is:** angle of trunk rotation of 7 degrees correlates with a Cobb angle of 20 degrees

# Scoliosis - Evaluation

- Do a careful physical examination with spinal examination
  - **Use of a scoliometer, cont:**
    - The operator-dependant measurements can include the following **sources of error:**
      - Inconsistent forward bend test **performance**,
      - Inconsistent **localization** of the scoliotic curve's apex
      - The scoliometer degree markings may be **poorly visualized** due to the size of the deviating ball
    - **Inter-rater** error for:
      - Thoracic curves – 2 degrees
      - Lumbar curves – 2.2 degrees
    - **Intra-rater** error for:
      - Thoracic curves – 1.2 degrees
      - Lumbar curves – 1.6 degrees

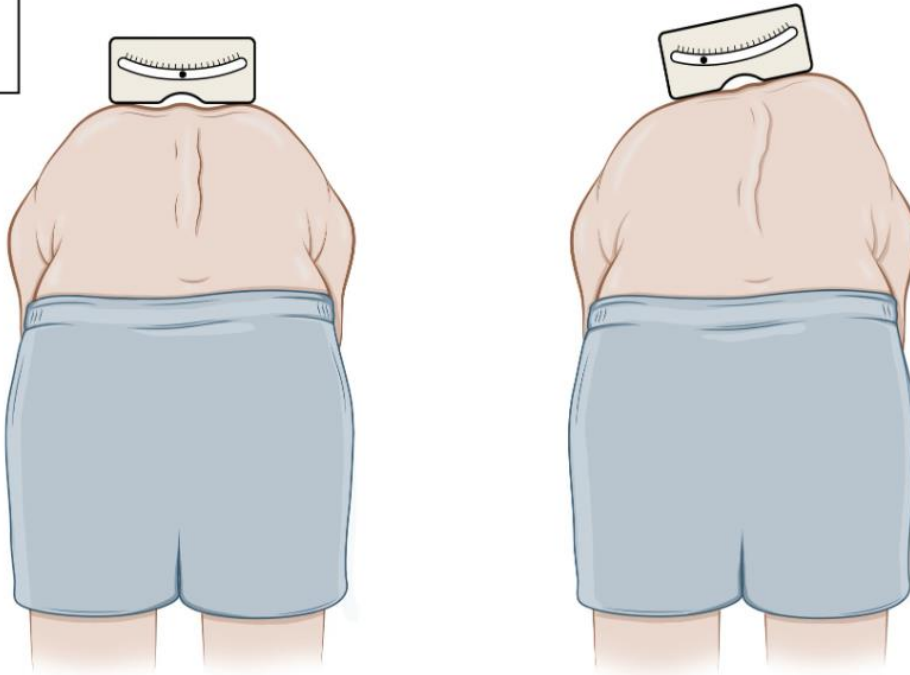
# Scoliosis – Evaluation Scoliometer

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Scoliometer



Normal

Abnormal

<https://www.childrenshospital.org/conditions-treatments/scoliosis>





# Use of the scoliometer



## Use of the scoliometer

The scoliometer is run along the patient's spine from cephalad to caudad while the patient is in the position assumed for the Adam's forward bend test. In the above photograph, the right thoracic prominence causes the right side of the scoliometer to deviate upward and the ball to deviate to the left. The angle of trunk rotation can also be measured with a scoliometer app on a smartphone.

Scherl, Susan, MD, et. al. Adolescent idiopathic scoliosis: Clinical features, evaluation, and diagnosis, UpToDate: literature review current through June 2026, topic last updated June 18, 2026. LINK: [https://www.uptodate.com.arktos.nyit.edu/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search\\_result&selectedTitle=2~11&usage\\_type=default&display\\_rank=2](https://www.uptodate.com.arktos.nyit.edu/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search_result&selectedTitle=2~11&usage_type=default&display_rank=2)



# Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
  - Before sending patient, treat patient with OMT to get good spinal mobilization.
  - Looking for any bony pathology, sacral base unleveling, femoral head/ilic crest levelness, and any scoliotic curves – type and severity.
  - **Cobb Method**

# Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
  - Is done to **confirm** presence of scoliosis, to **evaluate** the **type** (congenital, neuromuscular, idiopathic) and **severity**, and to **assess the skeletal maturity**, in relation to risk for progression.

# Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
  - **Indications:**
    - **Scoliometer reading  $>$  or  $=$  to 7 degrees** (may need treatment)
    - Presence of an **obvious spinal curve**
    - Thoracic or lumbar **asymmetry** seen on physical examination **in skeletally immature children, + family history of scoliosis**
    - To **monitor progression** in patients already diagnosed with AIS

# Scoliosis – Radiographic Evaluation

- Radiology – **standardized standing postural x-ray images**
  - **Initial radiographic evaluation for scoliosis:**
    - Standing, full length
    - Posteroanterior (PA) and lateral views
      - The PA view minimizes radiation to breasts and thyroid
      - As if looking at patient from behind, patient's heart is to the examiner's left
    - From C7 to the sacrum and iliac crest
    - All exposed on a single long film cassette (send patient to a facility where this is done, versus obtaining suboptimal x-rays)

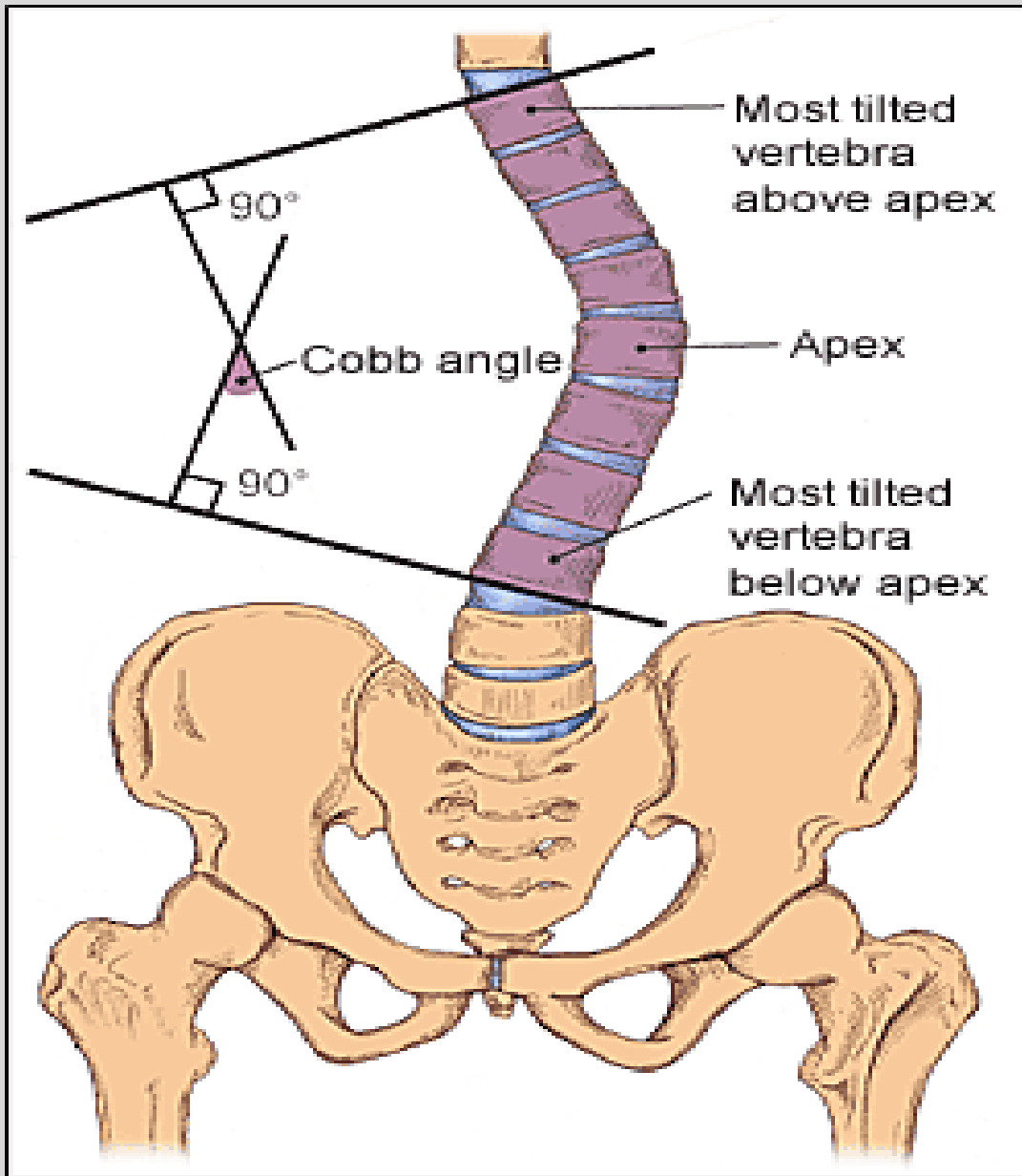
# Scoliosis – Radiographic Evaluation

- Radiology – **standardized *standing* postural x-ray images**
- **Initial radiographic evaluation for scoliosis, cont.:**
  - DO NOT have radiographs taken in the *supine* position, since this decreases the effect of gravity on the spinal curves, leading to underestimation of the spinal curve(s)
    - ***Look for air-fluid level in the stomach, which confirms that x-ray was done in standing position***
  - For a true leg length discrepancy, place the appropriate size block/lift under the shorter leg before the radiographic procedure.
    - ***To determine the pure scoliotic curve, without the additional compensation added in this case***
  - Lateral bending films
    - ***These are to be obtained by the operating surgeon as necessary, and can show the potential for surgical correction***

# Scoliosis – Radiographic Evaluation

- **Cobb Method**

- A line at the superior border of the highest vertebra inclined towards the concavity, and a line perpendicular to this.
- A line at the inferior border of the lowest vertebra inclined towards the concavity, and a line perpendicular to this.
- At the intersection of the two perpendicular lines, the degrees measured here give the Cobb Angle.



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### Thoracolumbar scoliosis

Posteroanterior radiograph demonstrating right thoracolumbar scoliosis. The air-fluid level in the stomach confirms that the radiograph was taken with the patient standing. Scoliosis radiographs are viewed with the patient's heart on the examiner's left (as if the examiner is standing behind the patient).

Scherl, Susan, MD, et. al. Adolescent idiopathic scoliosis: Clinical features, evaluation, and diagnosis, UpToDate: literature review current through June 2026, topic last updated June 18, 2026. LINK: [https://login.nyit.idm.oclc.org/login?url=https://www.uptodate-com.nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search\\_result&selectedTitle=2%7E11&usage\\_type=default&display\\_rank=2](https://login.nyit.idm.oclc.org/login?url=https://www.uptodate-com.nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search_result&selectedTitle=2%7E11&usage_type=default&display_rank=2)

# Scoliosis – Radiographic Evaluation

The scoliotic curve's **LOCATION** is defined by the “**Apical Vertebra,**” which is the vertebra that is most deviated and rotated from midline:

- Cervical – C2 to C6
- Cervicothoracic – C7 to T1
- Thoracic – T2 to T11
- Thoracolumbar – T12 to L1
- Lumbar – L2 to L4
- Lumbosacral – L5 or below

A common AIS curve pattern is a **Double Curve**, which is **right thoracic** and **left lumbar**, however, remember that there are many curve patterns/configurations that can happen.



# Scoliosis – Radiographic Evaluation

- Do the x-rays **suggest** any **underlying** congenital or neuromuscular etiology?
  - Paraspinal masses
  - Hemivertebrae or wedged vertebrae (congenital scoliosis)
  - Vertebral body lucency or erosion of the pedicle
    - Potentially - bone tumor

# Scoliosis – Radiographic Evaluation

- For **bone age/skeletal maturity** in adolescents with a ***rapidly progressing curve***:
  - Check x-ray of iliac apophysis (***Risser sign***)
  - Check x-ray of hand (***Sanders Skeletal Maturity Staging System***)
  - ***Also, can check the following (FYI):***
    - Check x-ray of ***epiphyses of distal radius, ulna, and small bones of the hand*** (the method of ***Tanner and Whitehouse***)
    - Check x-ray of ***elbow*** (assessment of ossification centers in the elbow)
      - ***Sauvegrain method***
      - The (simpler) ***olecranon*** method

# Scoliosis – Radiographic Evaluation

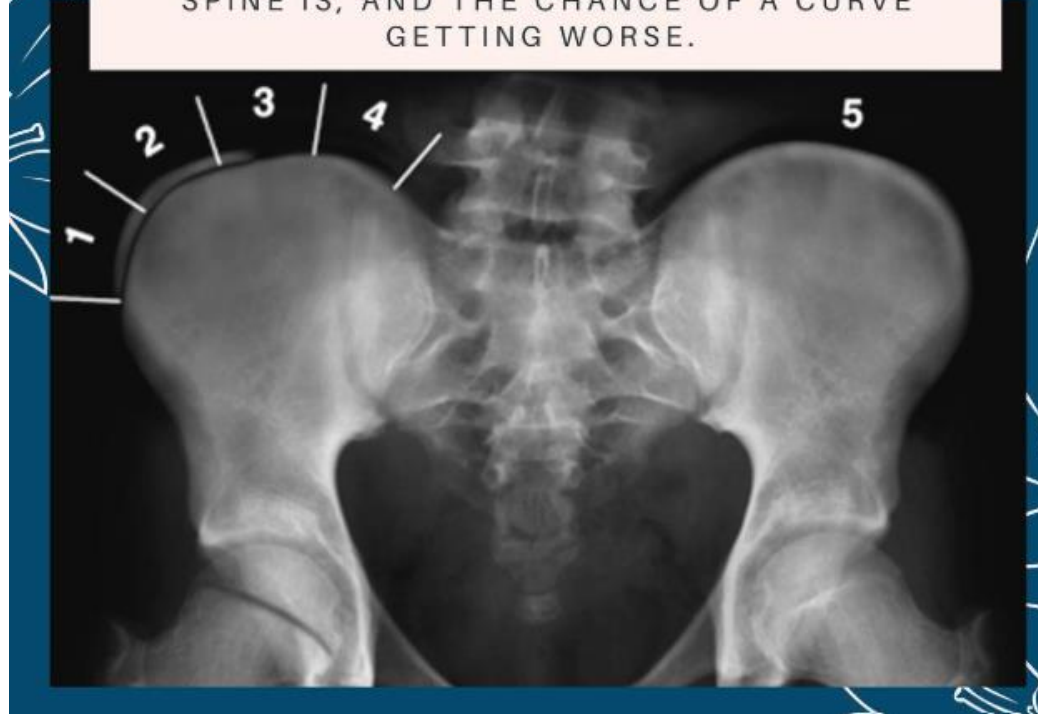
- For Bone age/skeletal maturity: x-ray of iliac apophysis (**Risser sign**):
  - *The Risser sign is a visual grading of the degree of ossification and/or fusion present in the iliac apophysis:*
    - 0 – no ossification
    - 1 – up to 25% ossification (Risser 1 usually occurs AFTER peak height velocity)
    - 2 – 26% to 50% ossification
    - 3 – 51% to 75% ossification
    - 4 – greater than 76% ossification
    - 5 – Full bony fusion of the apophysis
  - **A lower Risser grade, means that the patient has quite a bit more growth remaining, and is therefore at a greater risk for a scoliotic curve progression**

# Scoliosis – Radiographic Evaluation

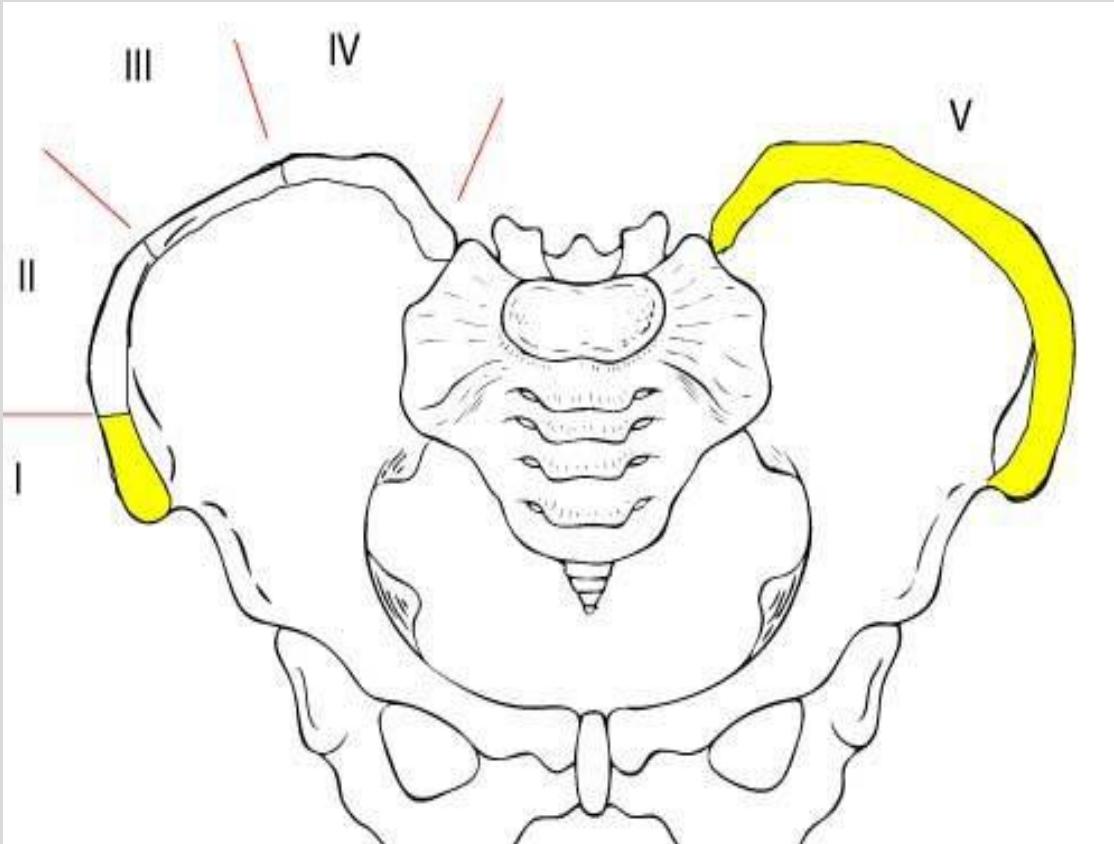
## Risser Sign

### Risser Sign

A RISSER SIGN IS THE THIN WHITE LINE ON TOP OF THE PELVIS, SEEN ON X-RAYS. IT DETERMINES HOW MUCH GROWTH SOMEONE HAS LEFT, HOW FLEXIBLE THE SPINE IS, AND THE CHANCE OF A CURVE GETTING WORSE.



# Risser sign for skeletal maturity



“The iliac apophysis ossifies in a stepwise fashion from anterolateral to posteromedial along the iliac crest. The Risser sign is a visual grading of the degree to which the iliac apophysis has undergone ossification and fusion; it is used to assess skeletal maturity. Risser grade 0 corresponds to no ossification. As depicted above, grade 1 describes up to 25 percent ossification; grade 2 describes 26 to 50 percent ossification; grade 3 describes 51 to 75 percent ossification; grade 4 describes greater than 76 percent ossification; and grade 5 describes full bony fusion of the apophysis.”

- Scherl, Susan, MD, et. al. Adolescent idiopathic scoliosis: Clinical features, evaluation, and diagnosis, UpToDate: literature review current through June 2026, topic last updated June 18, 2026. LINK: [https://www.uptodate.com.arktos.nyit.edu/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search\\_result&selectedTitle=2~11&usage\\_type=default&display\\_rank=2](https://www.uptodate.com.arktos.nyit.edu/contents/adolescent-idiopathic-scoliosis-clinical-features-evaluation-and-diagnosis?search=adolescent%20idiopathic%20scoliosis&source=search_result&selectedTitle=2~11&usage_type=default&display_rank=2)

The 5 grades of the Risser scale. Source: Department of Orthopaedic Surgery at University Stellenbosch, South Africa. LINK: <https://clear-institute.org/blog/risk-scoliosis-progressing/>

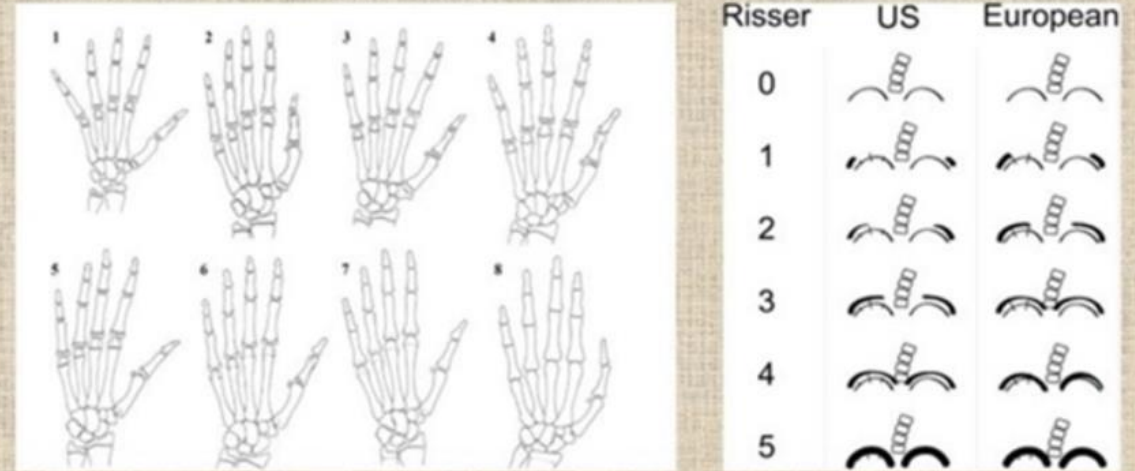
# Sanders Skeletal Maturity Staging System

- The Sanders Staging System is preferred by many practitioners.
- Compared to the Risser Sign, it may correlate better with peak height velocity.
- May predict the progression of scoliosis better than Risser Sign approach. Sanders Stages:
  - 3 or less = substantial growth remains
  - 4-7 = little remaining growth
  - 7-8 = mature skeleton

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Department of Osteopathic Manipulative Medicine

## Sanders stages vs Risser sign



| Sanders Stage                | Radiographic Features <sup>9</sup>                                     | Risser Stage | Radiographic Features                                                  |
|------------------------------|------------------------------------------------------------------------|--------------|------------------------------------------------------------------------|
| 1. Juvenile slow             | Digital epiphyses are not covered                                      | 0            | No ossification of the apophysis                                       |
| 2. Preadolescent slow        | All digital epiphyses are covered                                      | 0            | No ossification of the apophysis                                       |
| 3. Adolescent rapid (early)  | Most digits are capped. 2nd-5th MC epiphyses are wider than metaphyses | 0            | Triradiate cartilage open (peak height velocity)                       |
| 4. Adolescent rapid (late)   | Any of distal phalangeal physes are clearly beginning to close         | 0            | Triradiate cartilage remains (open growth plates in the long bones)    |
| 5. Adolescent steady (early) | All distal phalangeal physes are closed, while others are open         | 0            | Triradiate cartilage closed (menarche in female patients)              |
| 6. Adolescent steady (late)  | Middle or proximal phalangeal physes are closing                       | ≥ 1          | Ossification of the iliac apophysis ranges from 25% to 75%             |
| 7. Early mature              | Only distal radial physis is open. MC physal scars may be present      | 4            | 100% ossification of the iliac wing, with no fusion to iliac crest     |
| 8. Mature                    | Distal radial physis is completely closed                              | 5            | Fusion of the iliac apophysis to the iliac crest (cessation of growth) |

MC indicates metacarpal.

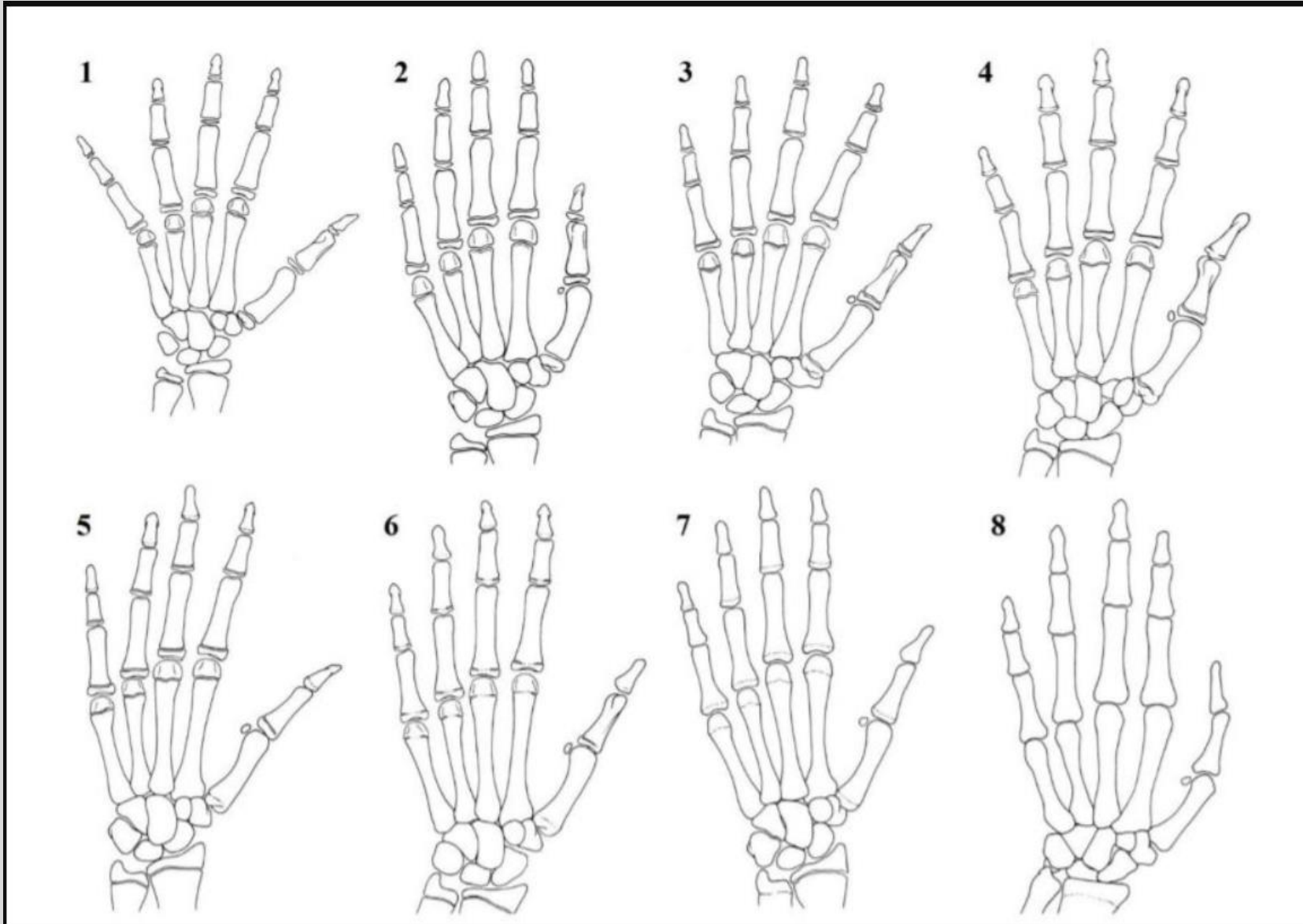
- <https://bracedforscoliosis.co.uk/2024/10/11/understanding-brace-weaning-for-scoliosis/>



# Sanders Staging System

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# Scoliosis – Radiographic Evaluation

- MRI indications:
  - Early onset before age 10, with rapid progression.
  - Patient complaints of significant pain associated with the scoliosis:
    - May also need CT or bone scan to rule out bone tumors or infection



# Scoliosis – Radiographic Evaluation

- MRI indications, cont.:
  - Associated neurologic s/s:
    - Headache, neck pain, absence of abdominal reflex, asymmetric lower extremity atrophy, pes cavus, midline skin lesions (e.g. vascular, pigmentary, hair patch)
  - Abnormal findings on the plain radiographs:
    - Findings correlating with congenital scoliosis
    - Findings suggestive of intraspinal pathology
    - Increased kyphosis

# Scoliosis – Radiographic Evaluation

- **Indications for REFERRAL to Orthopedic specialist:**
  - **Angle of trunk rotation (scoliometer measurement) of  $\geq 7$  degrees in patients with BMI  $< 85^{\text{th}}$  percentile, and not able to obtain a Cobb angle measurement**
  - **Angle of trunk rotation (scoliometer measurement) of  $\geq 5$  degrees in patients with BMI  $\geq 85^{\text{th}}$  percentile, and not able to obtain a Cobb angle measurement**
  - **Cobb angle between 20 and 24 degrees in males or premenarchal girls aged 12 to 14.**
  - **Cobb angle  $\geq 25$  degrees in any patient**
  - **Progression of Cobb angle of more than 5 degrees in any patient**

# Scoliosis – Radiographic Evaluation

- **Indications for REFERRAL to Orthopedic specialist:**
  - At time of referral, be clear about **which method of measurement was used – Cobb angle or Scoliometer** – since the degree values represent quite disparate values, with variable risks for progression.
  - ***Orthopedic referral is not necessary for incidental findings of scoliosis on a chest x-ray, when the Cobb angle measure is < 10 degrees.***

# Scoliosis – Treatment

*As it is not proper to cure the eyes without the head, nor the head without the body, neither is it proper to cure the body without the soul. - Socrates*

# Scoliosis – Treatment

- Generally, **observation, bracing, and surgery** are the main options for treatment.
- Physical therapy, chiropractic treatment, electrical stimulation, and biofeedback have been shown to be ineffective.
- ***As always, individual patient concerns must be considered.***
  - Note- the terms "male" and "female" are used to describe the sex assigned at birth. The treatment data may not apply to transgender and gender-diverse youth who are receiving gender-affirming hormone therapy.

- Scherl, Susan, MD, et. al. Adolescent idiopathic scoliosis: management and prognosis, UpToDate: literature review current through June 2026, topic last updated July 25, 2025. LINK: [https://login.nyit.idm.oclc.org/login?url=https://www.uptodate-com.nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-management-and-prognosis?search=adolescent%20idiopathic%20scoliosis&source=search\\_result&selectedTitle=1%7E11&usage\\_type=default&display\\_rank=1](https://login.nyit.idm.oclc.org/login?url=https://www.uptodate-com.nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-management-and-prognosis?search=adolescent%20idiopathic%20scoliosis&source=search_result&selectedTitle=1%7E11&usage_type=default&display_rank=1)

# Scoliosis – Treatment

- **Mild Scoliosis** – OMT, (Konstancin, per Campbell's Orthopedics) exercises, functional orthotics, education
- **Moderate Scoliosis** – OMT, exercises, education, bracing.
  - NOTE: For Mild and Moderate Scoliosis, **particular Isometric Yoga Poses OR the Schroth Method for Scoliosis** (*see CPG for Supplemental information, not required*) could be additional options considered, sometimes in conjunction with bracing, and the other options, as well.
- **Severe Scoliosis** – Surgery and adjunctive treatment as above.

# Scoliosis – Treatment

- OMT goal in the treatment of structural scoliosis is to **optimize function in the existing structure, not to straighten the curves.**
- Indirect treatment, fascial release, soft tissue treatment, gentle Balanced Ligamentous/Membranous Tension (BLT, BMT) treatments, to increase general body ROM.
- Abdominal and Psoas muscle strengthening.
- Can treat joint somatic dysfunctions as well.

# Scoliosis – Proposed Adjunct Treatment

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## Two Isometric Yoga Poses Reduce the Curves in Degenerative and Adolescent Idiopathic Scoliosis

Fishman, Loren M. MD, BPhil (Oxon); Groessl, Erik J. PhD; Bernstein, Paul OMS II

Topics in Geriatric Rehabilitation: October/December 2017 - Volume 33 - Issue 4 - p 231–237

doi: 10.1097/TGR.0000000000000159

Thieves' Market

**Abstract** **Author Information** **Article Outline**

**Objective:** Efficacy of 2 yoga poses for degenerative scoliosis (DS) and adolescent idiopathic scoliosis (AIS).

**Subjects and Methods:** Retrospective study of 49 DS and 25 AIS patients with initial Cobb angles less than 120° (DS) and less than 75° (AIS) performing side plank daily, with lumbar and thoracolumbar curves' convex side inferior. In complex curves, the half-moon pose was added. Outcome measure included Cobb angles after mean 5.8 and 9.4 months, respectively.

**Results:** Mean lumbar/thoracolumbar improvement: 23.7% ( $P < .001$ ) in DS, 34.2% ( $P = .001$ ) in AIS. Thoracic curves improved 27.6% ( $P = .001$ ) and 20.3% ( $P = .004$ ): 2.5% per month and 3.5% per month, respectively.

**Conclusions:** These yoga poses may reduce scoliosis in DS and AIS.

Columbia College of Physicians and Surgeons, New York, New York (Dr Fishman); Department of Family Medicine and Public Health, University of California San Diego, La Jolla (Dr Groessl); VA San Diego Healthcare System, San Diego, California (Dr Groessl); and Kentucky College of Osteopathic Medicine, Pikeville (Mr Bernstein).

**Correspondence:** Loren M. Fishman, MD, BPhil (Oxon), Columbia University Medical School, 133 East 58th St, New York, NY 10022 (Loren@sciatica.org).

Each patient and, where appropriate, her or his parents have given full consent for their information to be published in this paper in accordance with the Declaration of Helsinki updated at the WMA General Assembly in Fortaleza, Brazil, in October of 2013. The two people appearing in figures are not patients. They each have given full consent for these pictures to be shown in this publication.

**Level IV evidence:** Retrospective cohort and observational study with dramatic effect.

The authors declare that they have no conflicts of interest regarding this study.

Original Article

## Isometric Yoga-Like Maneuvers Improve Adolescent Idiopathic Scoliosis—A Nonrandomized Control Trial

Loren M Fishman, MD, BPhil (Oxon)<sup>1</sup> 

### Abstract

**Objective:** Assess therapeutic value of specific yoga poses for thoracic and lumbar adolescent idiopathic scoliosis (AIS) taught in office or Internet.

**Study Design:** Nonrandomized control trial: Fifty-six adolescents (mean age 14.0 years; mean Risser 3.0) were recruited from our clinic; 41 did the side-plank, the half-moon and elevated side plank poses as appropriate (treatment group) and 15 did not (controls). Thirty curves were treated in office, 30 via Internet. Curve change was evaluated by blinded serial Cobb angles, and analyzed using Mann-Whitney U, paired t-tests and  $\chi^2$ .

**Results:** Mean lumbar and thoracolumbar Cobb angle change was  $-9.2$  (95% CI =  $-11.8, -6.6$ ) in the treatment group and  $5.4$  (95% CI =  $1.7, 9.0$ ) in controls. Both treatment group improvement and deterioration in controls were significant (treatment group: paired t-test  $t = -7.1$ ,  $df = 40$ ,  $p = .000$ ; controls:  $t = 3.2$ ,  $df = 12$ ,  $p = .008$ ). Mean thoracic Cobb angle change was  $-7.1$  (95% CI =  $-13.1, -1.2$ ) in the treatment group and  $9.3$  (95% CI =  $4.5, 14.6$ ) in controls. Both changes were significant (paired t-test  $t = -3.3$ ,  $df = 21$ ,  $p = .022$  for treatment group;  $t = 4.5$ ,  $df = 5$ ,  $p = .006$  for controls). Nine Internet patients were non-compliant vs. 6 office patients. Office patients improved 1.6 degrees/month or 5.5%/month; Internet patients improved .72 degrees/month or 3.3%/month.

**Conclusion:** These yoga poses show promise for reversing adolescent idiopathic scoliosis. Telemedicine had greater non-compliance and lower efficacy but still produced patient improvement.

### Keywords

yoga, health care, young adults, scoliosis, telemedicine, complementary and alternative medicine

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# Scoliosis – Proposed Adjunct Treatment

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## Schroth physiotherapeutic scoliosis-specific exercises for adolescent idiopathic scoliosis: how many patients require treatment to prevent one deterioration? – results from a randomized controlled trial - “SOSORT 2017 Award Winner”

Sanja Schreiber<sup>1\*</sup>, Eric C Parent<sup>2</sup>, Doug L Hill<sup>3</sup>, Douglas M Hedden<sup>3</sup>, Marc J Moreau<sup>3</sup> and Sarah C Southon<sup>3</sup>

### Abstract

**Background:** Recent randomized controlled trials (RCTs) support using physiotherapeutic scoliosis-specific exercises (PSSE) for adolescents with idiopathic scoliosis (AIS). All RCTs reported statistically significant results favouring PSSE but none reported on clinical significance. The number needed to treat (NNT) helps determine if RCT results are clinically meaningful. The NNT is the number of patients that need to be treated to prevent one bad outcome in a given period. A low NNT suggests that a therapy has positive outcomes in most patients offered the therapy. The objective was to determine how many patients require Schroth PSSE added to standard care (observation or brace treatment) to prevent one progression (NNT) of the Largest Curve (LC) or Sum of Curves (SOC) beyond 5° and 10°, respectively over a 6-month interval.

**Methods:** This was a secondary analysis of a RCT. Fifty consecutive participants from a scoliosis clinic were randomized to the Schroth PSSE + standard of care group ( $n = 25$ ) or the standard of care group ( $n = 25$ ). We included males and females with AIS, age 10–18 years, all curve types, with curves 10°–45°, with or without brace, and all maturity levels. We excluded patients awaiting surgery, having had surgery, having completed brace treatment and with other scoliosis diagnoses. The local ethics review board approved the study (Pro00011552). The Schroth intervention consisted of weekly 1-h supervised Schroth PSSE sessions and a daily home program delivered over six months in addition to the standard of care. A prescription algorithm was used to determine which exercises patients were to perform. Controls received only standard of care. Cobb angles were measured using a semi-automatic system from posterior-anterior standing radiographs at baseline and 6 months.

We calculated absolute risk reduction (ARR) and relative risk reduction (RRR). The NNT was calculated as:  $NNT = 1 / ARR$ . Patients with missing values (PSSE group;  $n = 2$  and controls;  $n = 4$ ) were assumed to have had curve progression (worst case scenario). The RRR is calculated as  $RRR = ARR / CER$

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**Results:** For LC,  $NNT = 3.6$  (95% CI 2.0–28.2), and for SOC,  $NNT = 3.1$  (95% CI 1.9–14.2). The corresponding ARR was 28% for LC and 32% for the SOC. The RRR was 70% for LC and 73% for the SOC. Patients with complete follow-up attended 85% of prescribed visits and completed 82.5% of the home program. Assuming zero compliance after dropout, 76% of visits were attended and 73% of the prescribed home exercises were completed.

**Conclusions:** The short term of Schroth PSSE intervention added to standard care provided a large benefit as compared to standard care alone. Four (LC and SOC) patients require treatment for the additional benefit of a 6-month long Schroth intervention to be observed beyond the standard of care in at least one patient.

**Trial registration:** NCT01610908 April 2, 2012

**Keywords:** Physiotherapeutic scoliosis specific exercises, Number needed to treat, Clinical significance, Schroth, Exercise, Radiography, Cobb angle, Scoliosis, Adolescents, Spinal curvatures

<https://pmc-ncbi-nlm-nih-gov.nyit.idm.oclc.org/articles/PMC5684768/>

# Scoliosis – Treatment

- For **Adolescent Idiopathic Scoliosis (AIS)**, **the goal** is spinal curve at **skeletal maturity** with a **Cobb angle of 40 degrees or less.**
- If the **Cobb angle at skeletal maturity is 50 degrees or greater**, then these can progress at a rate of **1 degree per year after growth ceases.**
  - E.g. after 30 years → 30 degrees + 50 degrees = 80 degree+ curve.  
Association with restrictive lung disease, compromised pulmonary function.

# Scoliosis – Treatment

| Cobb angle           | Risser sign | Sanders Stage | Treatment recommendation                          |
|----------------------|-------------|---------------|---------------------------------------------------|
| 11-19 degrees        | n/a         | n/a           | Observation                                       |
| 20 to 24 degrees     | 0 to 2      | 1 to 3        | Follow closely                                    |
| 25 to 39 degrees     | 0 to 2      | 1 to 3        | Bracing                                           |
| 40 to 49 degrees     | 0 to 2      | 1 to 3        | Bracing, may consider<br>Surgery above 45 degrees |
| > or = to 50 degrees | Any stage   | Any stage     | Surgery                                           |

# Scoliosis – Treatment

- **Bracing:**

- Indicated for **skeletally immature** patients (**Risser 0 to 2, Sanders Stage 1 to 3**).
  - Not indicated for skeletally mature patients (Risser 3 or higher, Sanders Stage 4 to 8), as they have little growth remaining.
- **Cobb angle is 25 to 49 degrees at the time of presentation**
- **Cobb angle progresses  $>$  or  $=$  to 5 degrees in any 6-8 month observation period**
- **May prevent curve progression**, potentially reducing the need for surgery
- **Does not correct the curve present at the time of diagnosis**

# Scoliosis – Treatment

- **Bracing:**
  - **Contraindicated in patients who have reached skeletal maturity (*Risser 4 to 5, ring apophyses fused*)**
  - **Contraindicated in patients with a Cobb angle  $>$  or  $=$  to 50 degrees**
  - **Relatively contraindicated in patient's with a thoracic lordosis**

# Scoliosis – Treatment

- **Braces**

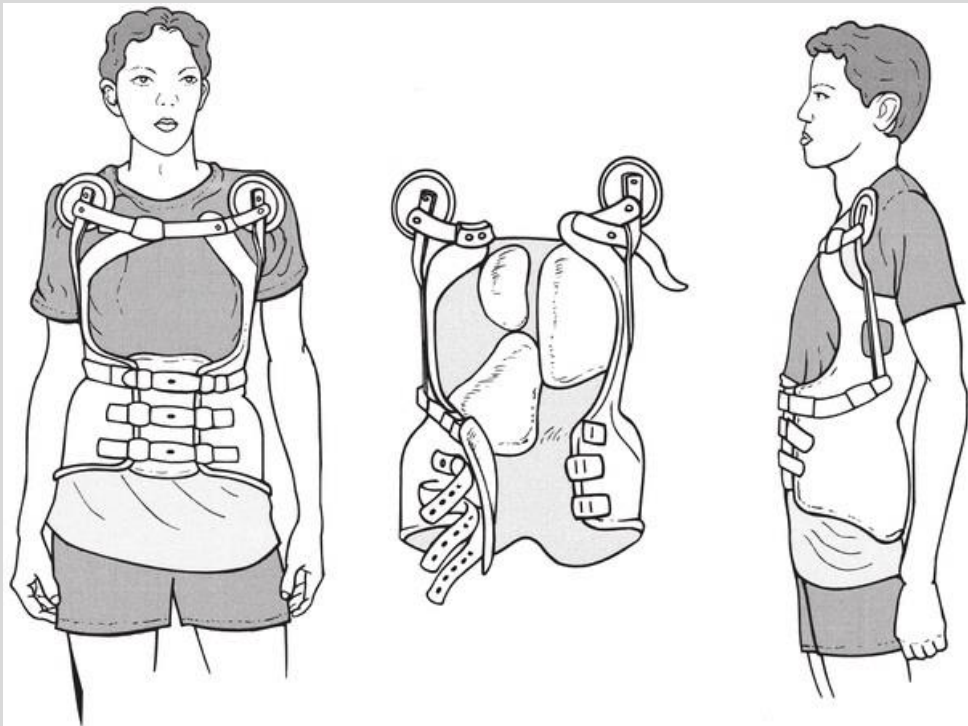
- **Underarm orthoses** utilized for most curves– **TLSOs (thoraco-lumbar-sacral orthosis)** like the **Boston Brace** are used now.
  - Also: the **Wilmington plastic jacket**, the **Charleston brace**, and the **Providence brace**
  - Prevents progression of larger curves when worn 18 hours/day.
- **Milwaukee Brace** (23 hours/day) **rarely used now**, a.k.a. **CTL SO (cervico-thoraco-lumbar-sacral orthosis, has an under-chin extension)**
  - 50% improvement after 6 months in the brace; but at 5 year follow-up, curve was the same as at diagnosis.

# Scoliosis – Treatment

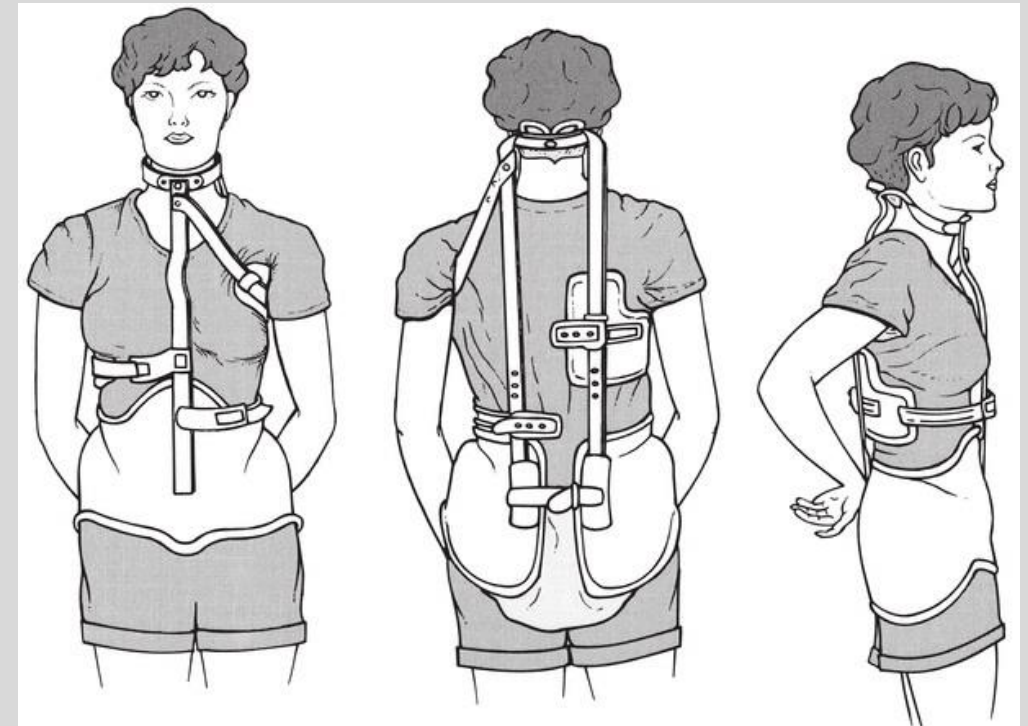
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## Boston Brace



## Milwaukee Brace





# Scoliosis – Treatment

- **Efficacy of bracing – *mixed results*** reported in various reviews of the literature.
- National Institutes of Health “**BrAIST**” trial.
  - The Bracing in Adolescent Idiopathic Scoliosis Trial (BrAIST)
  - Five-year, multicenter (20), prospective, randomized trial: compared bracing and observation in skeletally immature patients
  - Cobb angle between 25 to 40 degrees



# Scoliosis – Treatment

## Efficacy of bracing:

- Bracing treatment is **more effective with *increased hours* of daily wear:**

| Hours of wear<br>per day | Weighted mean<br>proportion of<br>success |
|--------------------------|-------------------------------------------|
| 23                       | 0.93                                      |
| 16                       | 0.62                                      |
| 8                        | 0.60                                      |

- Bracing treatment **more effective in females** than in males
- Bracing treatment **less effective in overweight patients**

# Scoliosis – Treatment

- **Bracing:**

- **Recommended brace wear of 23 hours per day in the past. Now, 18 hours per day is recommended, however-**
  - *BrAIST trial found that wearing the brace approx. 13 hours per day was effective.*
    - *With the exception of patients at Risser Stage 0, who may benefit from more hours wearing the brace.*
  - **Actual usage**, compared to the prior 23 hours that was recommended, **was approximately 65% of prescribed time**
  - Adherence to wearing the brace can be **monitored objectively** with **temperature or pressure sensors** inside the brace
- **Patient may wear the brace during sports; may remove for swimming, gymnastics-type sports as needed**

# Scoliosis – Treatment

- **Bracing:**

- **Check for proper fit of the brace, perform an in-brace radiograph**
  - **Look for the amount of curve correction, goal is 20% reduction of the in-brace Cobb angle → leads to greater chance of success**
  - **Brace treatment is continued until the end of the growth period;**
    - **Girls:** Two years post-menarche, Risser grade 4 or 5
    - **Boys:** Risser grade 5
    - Can either **stop completely, or wean down** to just nighttime usage (orthopedic practitioner clinical choice)
    - **Follow up at six-months post-bracing, and then yearly for a few years.** An initial curve collapse (several degrees) will soon stabilize.

# Scoliosis – Treatment

- **Bracing:**
  - **Monitor the brace treatment every 6-8 months both clinically and with follow up radiographs.**
    - **Out of brace radiographs are performed to check for any curve progression**
    - **Progression of the Cobb angle by  $>$  or  $=$  to 5 degrees during bracing treatment is a poor prognosis for the efficacy of this treatment for the patient, and surgery may be needed in these cases.**

# Scoliosis – Treatment

- **Surgery for AIS:**

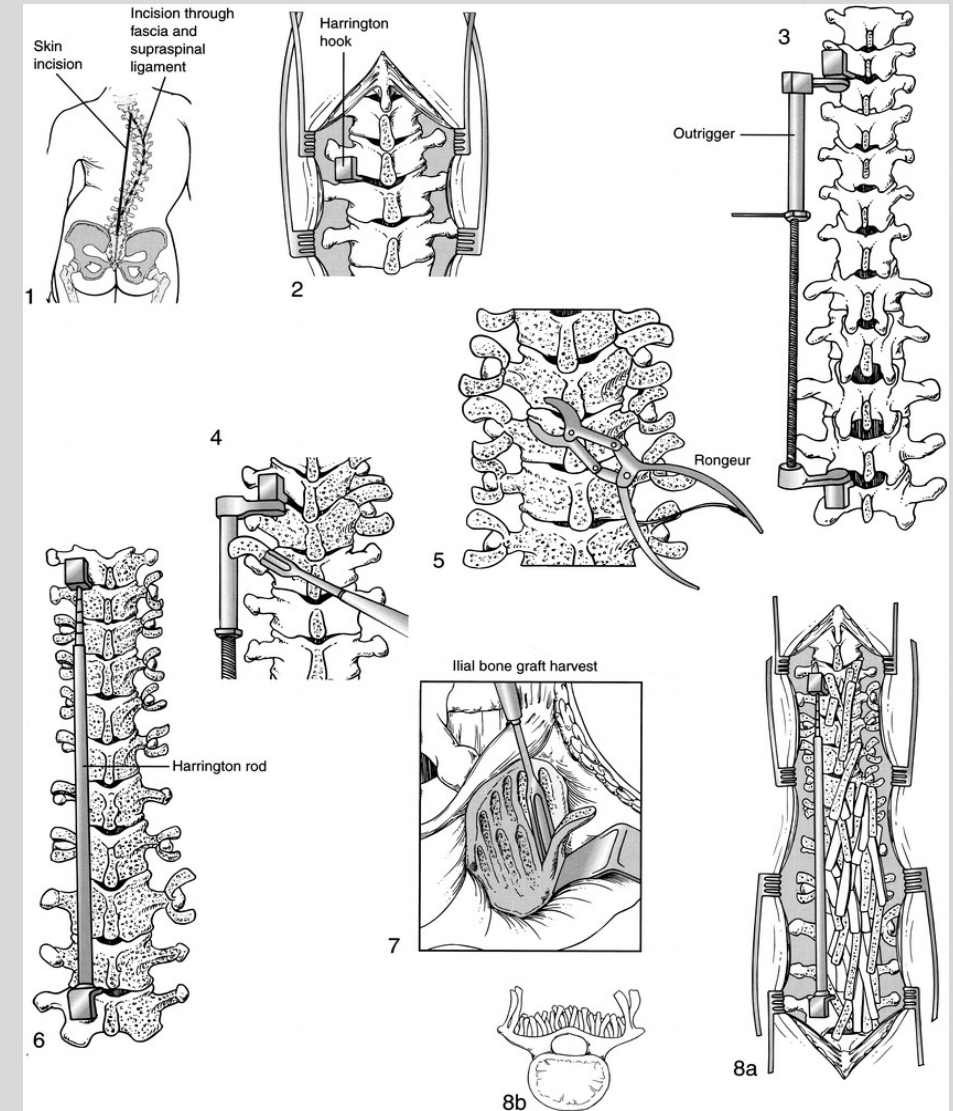
- **Primary goal** = utilizing **spinal fusion** for **prevention of curve progression**
- **Secondary goal** = **partial curve correction**, this is frequently achieved
- **Indications:**
  - **Skeletally immature patients**
  - **At diagnosis, patient has curve with Cobb angle of  $>$  or  $=$  to 50 degrees.**
    - **Sometimes** skeletally immature patients **with Cobb angle between 40 to 50 degrees *may* need surgery.**

# Scoliosis – Treatment

- **Surgery for AIS:**
  - Surgical procedures **utilized:**
    - **Posterior spinal fusion (most common)-**
      - Bone grafting and instrumentation
      - **Instrumentation has evolved** from the Harrington rods (cephalad and caudad spinal connections with hooks), to the modern segmental instrumentation (hooks, screws, wires, rods).
      - Modern instrumentation **provides the orthopedic surgeon with more control for positioning of the spine.**
      - Most important part of the surgery is obtaining **bony fusion**
        - **Autograft or allograft**

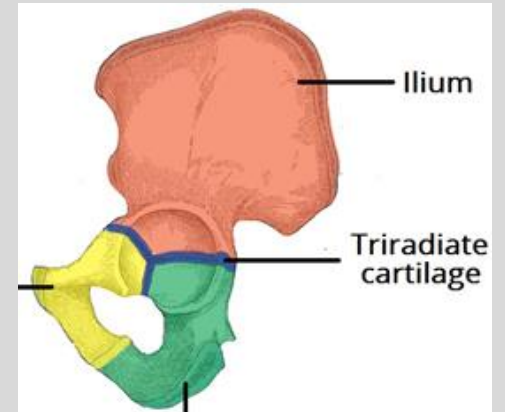
# Scoliosis – Treatment

Surgical fusion with **Harrington rod implantation**. This extensive surgery is reserved for patients with curves greater than 45 degrees. A major clinical goal is to **stop structural progression of a curve** before it seriously compromises cardiopulmonary function.



From: Chapter 29 Postural Considerations in Osteopathic Diagnosis and Treatment, Figure 29.34, Foundations of Osteopathic Medicine: Philosophy, Science, Clinical Applications, and Research, 4e, 2018, LINK:  
<https://login.nyit.idm.oclc.org/login?url=https://osteopathicmedicine-lwwhealthlibrary-com.nyit.idm.oclc.org/content.aspx?sectionid=209547150&bookid=2582>

# Scoliosis – Treatment



<https://teachmeanatomy.info/pelvis/bones/hip-bone/>

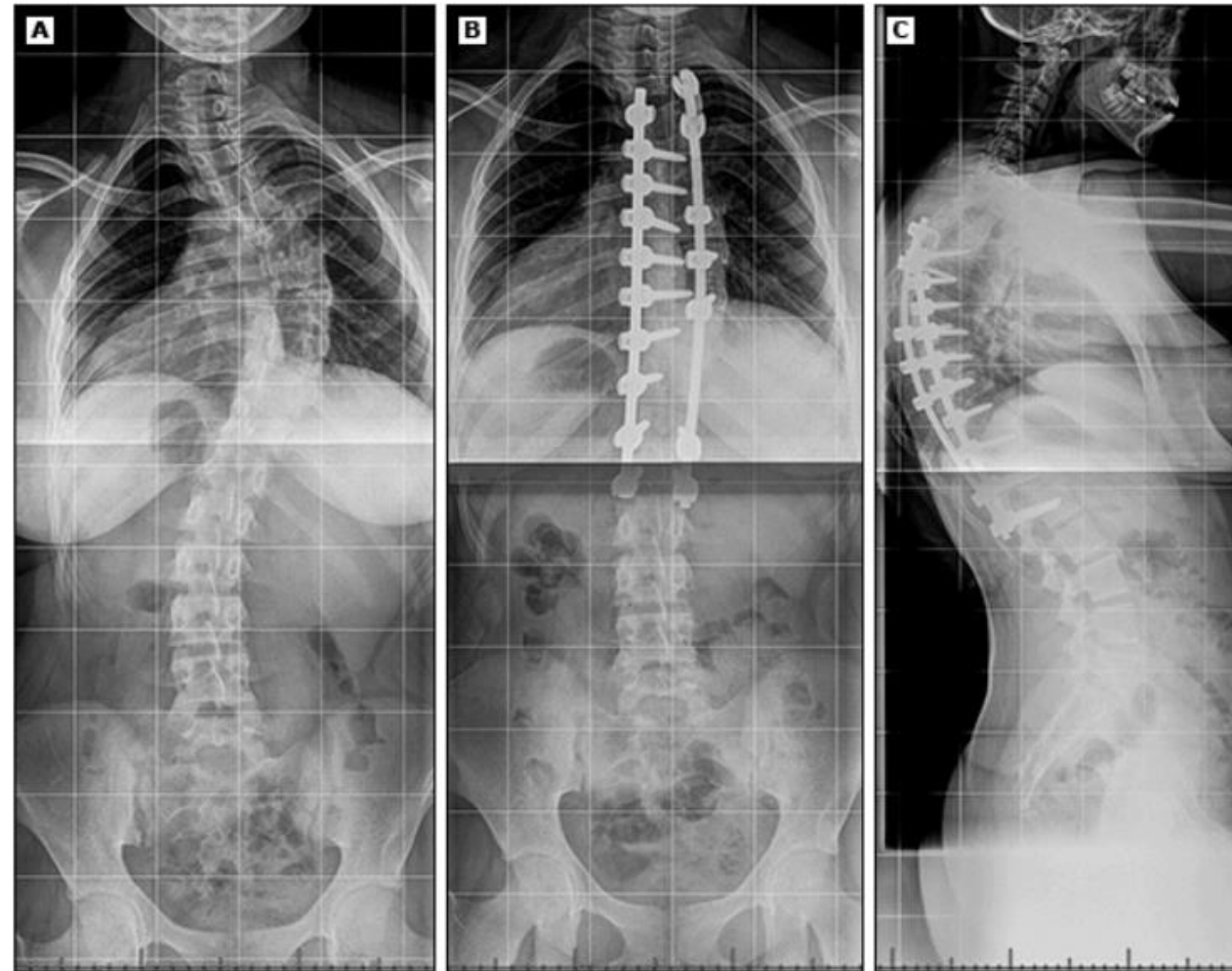
- **Surgery for AIS:**

- Surgical procedures **utilized**, continued:
  - **Posterior spinal fusion *with* Anterior spinal fusion:**
    - This is sometimes done in skeletally immature patients with open **triradiate cartilages**.
    - Goal is to prevent the occurrence of **the “crankshaft phenomenon”**
      - This phenomenon occurs after a *posterior* spinal fusion, where the anterior spine continues to grow.
      - This *will cause a severe alignment deformity, both rotationally and in the sagittal plane*
      - Some **evidence** that the modern segmental type of instrumentation obviates the need for an anterior spinal fusion



# Scoliosis – Treatment

## Posterior spinal fusion



“Preoperative posteroanterior (PA) and postoperative PA and lateral radiographs (panels A, B, and C, respectively) of a patient who underwent posterior spinal fusion and segmental spinal instrumentation with pedicle screws. Note on the postoperative lateral how the use of segmental instrumentation allows contoured rods to be attached to the spine to better maintain normal physiologic sagittal plane curves (thoracic kyphosis and lumbar lordosis). By orthopedic convention, scoliosis images are viewed as if looking from the patient's back; the left side of the image is the left side of the patient.”

*Courtesy of William Phillips, MD.*

Scherl, Susan, MD, et. al. Adolescent idiopathic scoliosis: management and prognosis, UpToDate: literature review current through June 2026, topic last updated July 25, 2025. LINK: [https://login.nyit.idm.oclc.org/login?url=https://www.uptodate.com.nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-management-and-prognosis?search=adolescent%20idiopathic%20scoliosis&source=search\\_result&selectedTitle=1%7E11&usage\\_type=default&display\\_rank=1](https://login.nyit.idm.oclc.org/login?url=https://www.uptodate.com.nyit.idm.oclc.org/contents/adolescent-idiopathic-scoliosis-management-and-prognosis?search=adolescent%20idiopathic%20scoliosis&source=search_result&selectedTitle=1%7E11&usage_type=default&display_rank=1)

# Scoliosis – Treatment

- **Surgery for AIS:**

- Surgical procedures, **outcomes:**
  - **95% of cases have successful fusion**
    - **Complications:** implant failure, pseudoarthrosis (failure of fusion), neurologic injury, infection, blood loss
    - **Intraoperative monitoring advances have further decreased the risk of complications**
  - ***Limited evidence*** regarding the ***primary goal of prevention of curve progression*** with the spinal fusion procedure
  - ***The secondary goal of coronal curve correction had variable results*** between 48-67% correction with a posterior fusion, and 71-93% with anterior fusion surgical procedures.

# Scoliosis – Treatment

- **Surgery for AIS:**

- *Post-operatively, for 9-12 months, sports participation is not recommended.*
  - Once spine has fused, sports are permitted, *except for collision sports.*
- **Long-term follow up data not yet available** for patients who have had the **modern segmental instrumentation performed.**

# Scoliosis – Treatment

- **Curve progression in AIS:**
  - ***Untreated curves may continue to progress after skeletal maturity***
    - Occurs in 2/3 of patients
    - ***Curves <30 degrees are much less likely to progress***
    - One **50-year follow up** showed patients had **little impairment** other than ***back pain and cosmetic concerns***

# Scoliosis – Treatment

- **Curve progression in AIS, cont.:**
  - **Curves treated with bracing and surgery can also progress:**
    - **Bracing treatment**, patients followed for 22 years, average coronal curve progression of **7.9 degrees**
    - **Surgical treatment**, patients followed for 23 years, average coronal curve progression of **3.5 degrees**

# Scoliosis – Treatment

- **Scoliosis and back pain:**

- Whether the scoliosis is ***treated or untreated***, there is a **slightly increased occurrence of back pain in adults with AIS**, compared to controls.
  - There ***may be a correlation*** between the ***initial magnitude of the curve*** and ***later back pain complaints***
  - ***Patients with AIS treated with bracing or surgery may also have increased chance of degenerative disc changes.***

# Scoliosis – Treatment

- **Scoliosis and pregnancy:**

- A patient's pregnancy ***will not affect*** her scoliotic curve's progression
  - **The risk of progression not affected by:**
    - The number of pregnancies
    - Patient's age at the time of the first pregnancy
    - The curve stability
- The scoliotic curve itself ***does not seem to affect the outcome*** of the pregnancy
  - **No apparent increase in cesarean section deliveries or other health concerns in pregnancy**, as compared to controls.

# Scoliosis – Treatment: Summary

- **Generally, in growing patients with a flexible 20-30 degree curve, with documented progression of 5 degrees or more, orthotic treatment is indicated. Adjunctive treatments may be included.**
- **Larger curves of 30-40 degrees in growing children, found upon the initial visit, are treated with orthotics right away.**
- **40-50 degrees – surgery *or may consider orthotics for double major curves***
- **>50 degrees - surgery**



# Summary Slide

1. Understanding the various types and classifications of scoliosis and knowing the standard radiological diagnostic method for assessing severity of scoliosis.
2. Understanding the treatment options and outcomes for scoliosis.

# General References

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